DEPI
Digital Egypt Pioneers Initiative
Track: IND3-G2

ToxWatch

A safety-checked AI first answer for poisoned patients

Graduation project · n8n · synthetic data only

AI drafts · a clinician decides

1 Business Problem

- Poisoning decisions need speed
- No toxicologist at night (hypothesis)
- Chatbot answers: no sources
- 21,492 patients in one centre, 2019

2 Solution

- Cited, safety-checked draft
- Risk tier on every question
- Critical → human approval
- Identifiers masked first

3 Target Users

- Emergency doctors, residents
- Clinical pharmacists
- Buyer: hospital director (hypothesis)
- Patients: never see the system

4 Core Features

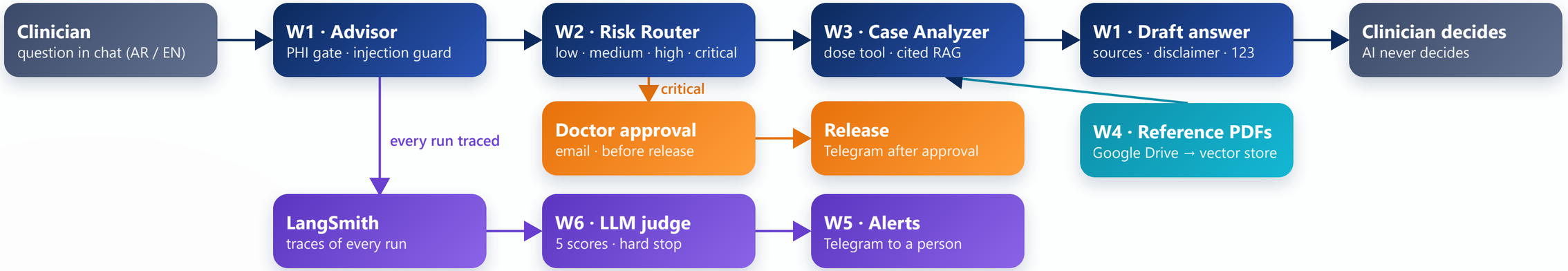
- PHI gate · injection guard
- Dose by tool, not model
- Cited answers (RAG)
- Audit · traces · alerts · hard stop

5 AI Components

- LLM agents (Gemini)
- Local Llama 3.2 · low risk
- RAG + rerank (Cohere)
- LLM-as-judge (5 scores)

System Architecture · six connected n8n workflows

Workflow · Knowledge · Human step · Monitoring



Technology Stack

- n8n
- Gemini
- Llama 3.2 · Ollama
- Cohere Rerank
- Presidio
- LangSmith
- Telegram
- Gmail
- Google Drive
- openFDA
- PubMed
- Docker

Development Timeline

- ✓ **Business Discovery** — Brief
- ✓ **Solution Design** — 6 spec cards
- ✓ **Build** — 6 workflows
- ✓ **Integration & Testing** — 3 scenarios · 12 cases
- ✓ **Business Pitch** — 10 slides

Project Outcomes

- ✓ 6 n8n workflows, published and running
- ✓ Test Report · 3 scenarios · 22 issues logged
- ✓ Business Brief · Solution Design · 5 docs
- ✓ 10 pitch slides · human scoring sheet

S1 · dose 10,000 mg

S2 · Arabic, no guess

S3 · held for a doctor, then approved

S1–S2: AI judge 5.0 · humans 4.0–4.6